

ZIMSKI ISPITNI ROK - RJEŠENJA

23.2.2025.

1. (a) $|z|^6 = |\bar{z}(1-i)|$
 $|z|^6 = |z|\sqrt{2}$

Napišemo z u polarnom obliku, $z = r(\cos(\varphi) + i\sin(\varphi))$. Tada r zadovoljava $r^6 = r\sqrt{2}$. Rješenja te jednačbe su $r = 0$ i $r = 2^{\frac{1}{10}}$. $\arg(z^6) = 6\arg(z)$ te $\arg(\bar{z}(1-i)) = \arg(\bar{z}) + \arg(1-i) + 2k\pi$, $\arg(\bar{z}) = -\arg(z)$, $\arg(1-i) = \frac{7\pi}{4}$. Stoga φ zadovoljava jednačbu $6\varphi = -\varphi + \frac{7\pi}{4} + 2k\pi$. Odnosno $\varphi = \frac{\pi}{4} + \frac{2k\pi}{7}$, $k = 0, 1, \dots, 6$. Dakle, rješenja su:

$$z_0 = 0$$

$$z_{k+1} = 2^{\frac{1}{10}} \left(\cos\left(\frac{\pi}{4} + \frac{2k\pi}{7}\right) + i \sin\left(\frac{\pi}{4} + \frac{2k\pi}{7}\right) \right), k = 0, 1, \dots, 6$$

(b) Neka je $x_1 < x_2$. Tada je $f(x_2) < f(x_1)$ jer je f strogo padajuća. g je strogo rastuća pa je $g \circ f(x_2) < g \circ f(x_1)$, odnosno $g \circ f$ je također strogo padajuća. Funkcija h je strogo monotona pa je ujedno i injekcija.

(c) $\operatorname{ch}(\ln x) = \frac{5}{2}$. Da bi x bio u domeni, mora vrijediti $x > 0$. Raspišimo $\operatorname{ch}(\ln(x))$ po definiciji:

$$\operatorname{ch}(\ln x) = \frac{e^{\ln x} + e^{-\ln x}}{2} = \frac{x + \frac{1}{x}}{2} = \frac{x^2 + 1}{2x}.$$

Početna jednačba se zato može napisati u obliku:

$$\frac{x^2 + 1}{2x} = \frac{5}{2}$$

Množenjem s x dobijemo slijedeću kvadratnu jednačbu:

$$x^2 - 5x + 1 = 0.$$

Rješenja te kvadratne jednačbe su $x_{1,2} = \frac{5 \pm \sqrt{21}}{2}$. Oba rješenja kvadratne jednačbe zadovoljavaju uvjet $x > 0$ pa su oba ujedno i rješenja početne jednačbe.

2. (a) Neka je $M > 0$, te $n_0 \in \mathbb{N}$ takav da je $n_0 > M^2$. Tada je za svaki $n \geq n_0$ $a_n = \sqrt{n+2} \geq \sqrt{n_0+2} \geq \sqrt{M^2+2} \geq \sqrt{M^2} = M$.

(b)

$$\begin{aligned} \lim_{n \rightarrow \infty} (a_{n+1} - a_n) &= \lim_{n \rightarrow \infty} (\sqrt{n+3} - \sqrt{n+2}) \\ &= \lim_{n \rightarrow \infty} ((\sqrt{n+3} - \sqrt{n+2}) \frac{\sqrt{n+3} + \sqrt{n+2}}{\sqrt{n+3} + \sqrt{n+2}}) \\ &= \lim_{n \rightarrow \infty} \frac{n+3 - (n+2)}{\sqrt{n+3} + \sqrt{n+2}} \\ &= \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+3} + \sqrt{n+2}} \\ &= 0 \end{aligned}$$

(c)

$$\begin{aligned}
\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} &= \lim_{n \rightarrow \infty} \frac{\sqrt{n+3}}{\sqrt{n+2}} \\
&= \lim_{n \rightarrow \infty} \frac{\sqrt{1 + \frac{3}{n}}}{\sqrt{1 + \frac{2}{n}}} \\
&= 1
\end{aligned}$$

3. (a)

$$\begin{aligned}
f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\ln^2(x+h) - \ln^2(x)}{h} \\
&= \lim_{h \rightarrow 0} \left[\frac{\ln(x+h) - \ln(x)}{h} \cdot (\ln(x+h) + \ln(x)) \right] \\
&= \lim_{h \rightarrow 0} \left[\frac{1}{h} \ln \frac{x+h}{x} \right] \cdot \lim_{h \rightarrow 0} (\ln(x+h) + \ln(x)) \\
&= \lim_{h \rightarrow 0} \ln \left(1 + \frac{h}{x} \right)^{1/h} \cdot \lim_{h \rightarrow 0} (\ln(x+h) + \ln(x)) = \ln e^{1/x} \cdot 2 \ln x = \frac{2 \ln x}{x}
\end{aligned}$$

(b)

$$\begin{aligned}
f'(x) &= (x \operatorname{tg} x + (x^2 + 1)^x)' = (x \operatorname{tg} x)' + \left(e^{x \ln(x^2 + 1)} \right)' \\
&= \operatorname{tg} x + \frac{x}{\cos^2 x} + e^{x \ln(x^2 + 1)} \left(\ln(x^2 + 1) + \frac{2x^2}{x^2 + 1} \right) \implies f'(0) = 0.
\end{aligned}$$

4. $f(x) = \operatorname{arctg} \frac{x^2}{x-2}$. Domena funkcije je $\langle -\infty, 2 \rangle \cup \langle 2, \infty \rangle$.

$$\begin{aligned}
\lim_{x \rightarrow 2^+} \operatorname{arctg} \frac{x^2}{x-2} &= \frac{\pi}{2} \\
\lim_{x \rightarrow 2^-} \operatorname{arctg} \frac{x^2}{x-2} &= -\frac{\pi}{2} \\
\lim_{x \rightarrow \infty} \operatorname{arctg} \frac{x^2}{x-2} &= \frac{\pi}{2} \\
\lim_{x \rightarrow -\infty} \operatorname{arctg} \frac{x^2}{x-2} &= -\frac{\pi}{2}
\end{aligned}$$

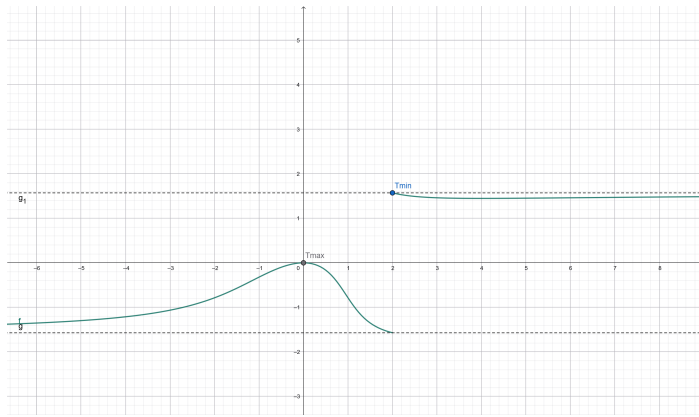
Derivacija funkcije f je:

$$\begin{aligned}
f'(x) &= \frac{1}{1 + \left(\frac{x^2}{x-2} \right)^2} \frac{2x(x-2) - x^2}{(x-2)^2} \\
&= \frac{x(x-4)}{(x-2)^2 + x^4}.
\end{aligned}$$

Stacionarne točke su $x = 0, x = 4$. Analizirajmo rast i pad funkcije:

	$\langle -\infty, 0 \rangle$	$\langle 0, 2 \rangle$	$\langle 2, 4 \rangle$	$\langle 4, \infty \rangle$
$f'(x)$	+	-	-	+
$f(x)$	$\nearrow$	$\searrow$	$\searrow$	$\nearrow$

Iz tablice vidimo da se lokalni maksimum postiže za $x = 0$ i iznosi 0 te lokalni maksimum za $x = 4$ te iznosi $\operatorname{arctg} 8$.



Slika 1: Graf funkcije

5. (a)

$$\begin{aligned}
 \frac{x^2}{x^4 - 1} &= \frac{x^2}{(x-1)(x+1)(x^2+1)} \\
 &= \frac{A}{x-1} + \frac{B}{x+1} + \frac{Cx+D}{x^2+1} \\
 &= \frac{A(x^2+1)(x+1) + B(x-1)(x^2+1) + (Cx+D)(x^2-1)}{x^4-1} \\
 &= \frac{(A+B+C)x^3 + (A-B+D)x^2 + (A+B-C)x + (A-B-D)}{x^4-1}
 \end{aligned}$$

Koeficijenti zadovoljavaju sustav linearnih jednadžbi:

$$A + B + C = 0$$

$$A - B + D = 1$$

$$A + B - C = 0$$

$$A - B - D = 0.$$

Rješenja tog sustava je $A = \frac{1}{4}, B = -\frac{1}{4}, C = 0, D = \frac{1}{2}$.

$$\begin{aligned}
 \int \frac{x^2}{x^4 - 1} dx &= \frac{1}{4} \int \frac{1}{(x-1)} dx - \frac{1}{4} \int \frac{1}{x+1} dx + \frac{1}{2} \int \frac{1}{x^2+1} dx \\
 &= \frac{1}{4} \ln|x-1| - \frac{1}{4} \ln|x+1| + \frac{1}{2} \arctg(x) + C, C \in \mathbb{R}
 \end{aligned}$$

(b)

$$\begin{aligned}
\int_0^1 x\sqrt{4-x^2} &= [x = 2 \sin u, dx = 2 \cos u du] = \int_0^{\frac{\pi}{6}} 8 \sin u |\cos u| \cos u du \\
&= \int_0^{\frac{\pi}{6}} 8 \sin u \cos^2 u du \\
&= [v = \cos u, dv = -\sin u du] = -8 \int_1^{\frac{\sqrt{3}}{2}} v^2 dv \\
&= -\frac{8}{3} v^3 \Big|_1^{\frac{\sqrt{3}}{2}} \\
&= \frac{8}{3} \left(1 - \frac{3\sqrt{3}}{8}\right) \\
&= \frac{8}{3} - \sqrt{3}.
\end{aligned}$$

6. (a) Integrand $f(x) = \frac{1}{\sqrt{1-x^6}}$ je neomeđen u okolini točke $x = 1$. S obzirom da vrijedi

$$\lim_{x \rightarrow 1^-} \frac{\frac{1}{\sqrt{1-x^6}}}{\frac{1}{\sqrt{1-x}}} = \frac{\sqrt{1-x}}{\sqrt{(1-x)(1+x+x^2+x^3+x^4+x^5)}} = \frac{1}{\sqrt{6}},$$

po usporednom kriteriju (T12.3.1):

$$\begin{aligned}
\int_0^1 \frac{1}{\sqrt{1-x^6}} dx &\sim \frac{1}{\sqrt{6}} \int_0^1 \frac{1}{\sqrt{1-x}} dx = \lim_{\epsilon \rightarrow 0} \frac{1}{\sqrt{6}} \int_0^{1-\epsilon} \frac{1}{\sqrt{1-x}} dx \\
&= \lim_{\epsilon \rightarrow 0} \frac{1}{\sqrt{6}} (-2\sqrt{1-x}) \Big|_{x=0}^{1-\epsilon} \\
&= \frac{1}{\sqrt{6}} \lim_{\epsilon \rightarrow 0} (-2\epsilon + 2) \\
&= \frac{2}{\sqrt{6}}.
\end{aligned}$$

Dakle, dani integral konvergira.

(b) Odredimo sjecišta krivulja:

$$\begin{aligned}
\frac{8}{x^2+4} &= \frac{x^2}{4} \\
x^4 + 4x^2 - 32 &= 0.
\end{aligned}$$

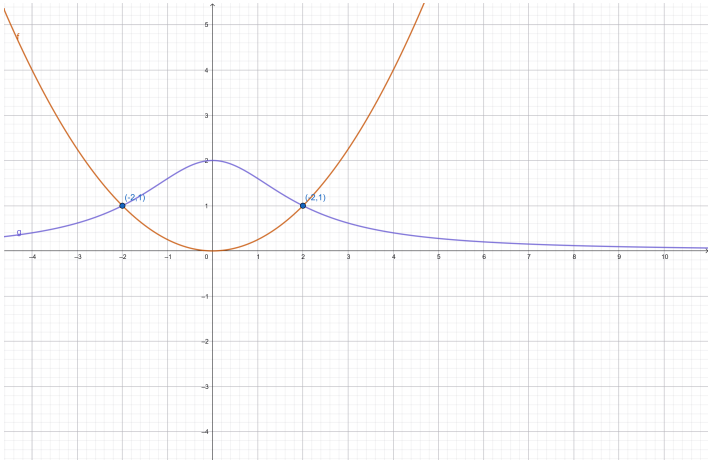
Uvedimo supstituciju $t = x^2$:

$$t^2 + 4t - 32 = 0.$$

Rješenja kvadratne jednadžbe su $t_1 = 4, t_2 = -8$, tj. $x_1 = -2, x_2 = 2$.

Skica lika je na slici 2. Traženi volumen je

$$\begin{aligned}
V &= 2\pi \int_0^2 x \cdot \frac{8}{x^2+4} dx - 2\pi \int_0^2 x \cdot \frac{x^2}{4} dx \\
&= [t = x^2 + 4, dt = 2x dx] = 2\pi \int_4^8 \frac{4}{t} dt - 2\pi \frac{1}{16} x^4 \Big|_{x=0}^{x=2} \\
&= 8\pi \ln |t| \Big|_{t=4}^{t=8} - 2\pi \\
&= 8\pi \ln 2 - 2\pi.
\end{aligned}$$



Slika 2: Skica